

ProjectManagerApp — Hướng Dẫn 06: FXML & CSS

Vị trí: `src/com/projectmanager/ui/views/*.fxml` · `src/com/projectmanager/ui/styles/main.css`

Lưu ý đường dẫn:

- Trong FXML: `stylesheets="@../styles/main.css"` → tương đối từ thư mục `views/`
- Trong SceneSwitcher: `App.class.getResource("/com/projectmanager/ui/views/" + fxml)` → classpath root

login.fxml

```
<?xml version="1.0" encoding="UTF-8"?>
<?import javafx.scene.layout.*?>
<?import javafx.scene.control.*?>
<?import javafx.geometry.*?>

<AnchorPane prefHeight="500.0" prefWidth="400.0"
    xmlns:fx="http://javafx.com/fxml/1"
    fx:controller="com.projectmanager.ui.controllers.LoginController"
    stylesheets="@../styles/main.css">

    <VBox alignment="CENTER" spacing="16"
        AnchorPane.leftAnchor="60" AnchorPane.rightAnchor="60"
        AnchorPane.topAnchor="70" AnchorPane.bottomAnchor="70">

        <Label text="Project Manager" styleClass="title"/>
        <Label text="Đăng Nhập" styleClass="subtitle"/>

        <VBox spacing="4">
            <Label text="Username" styleClass="field-label"/>
            <TextField fx:id="txtUsername" promptText="Nhập username"
styleClass="field"/>
        </VBox>

        <VBox spacing="4">
            <Label text="Mật Khẩu" styleClass="field-label"/>
            <PasswordField fx:id="txtPassword" promptText="Nhập mật khẩu"
styleClass="field"/>
        </VBox>

        <Button text="Đăng Nhập" onAction="#handleLogin"
            styleClass="btn-primary" maxWidth="Infinity"
defaultButton="true"/>

        <Label fx:id="lblMessage" wrapText="true" alignment="CENTER"
maxWidth="Infinity"/>

        <Separator opacity="0.3"/>
    </VBox>
</AnchorPane>
```

```

        <Button text="Tao Tai Khoan Moi" onAction="#goRegister"
              styleClass="btn-secondary" maxWidth="Infinity"/>
    </VBox>
</AnchorPane>

```

register.fxml

```

<?xml version="1.0" encoding="UTF-8"?>
<?import javafx.scene.layout.*?>
<?import javafx.scene.control.*?>
<?import javafx.geometry.*?>

<AnchorPane prefHeight="600.0" prefWidth="400.0"
  xmlns:fx="http://javafx.com/fxml/1"
  fx:controller="com.projectmanager.ui.controllers.RegisterController"
  stylesheets="@../styles/main.css">

    <VBox alignment="CENTER" spacing="12"
      AnchorPane.leftAnchor="60" AnchorPane.rightAnchor="60"
      AnchorPane.topAnchor="50" AnchorPane.bottomAnchor="50">

        <Label text="Project Manager" styleClass="title"/>
        <Label text="Tao Tai Khoan" styleClass="subtitle"/>

        <VBox spacing="4">
            <Label text="Username (it nhat 3 ky tu)" styleClass="field-label"/>
            <TextField fx:id="txtUsername" promptText="username"
styleClass="field"/>
        </VBox>

        <VBox spacing="4">
            <Label text="Mat Khau (it nhat 6 ky tu)" styleClass="field-label"/>
            <PasswordField fx:id="txtPassword" promptText="mat khau"
styleClass="field"/>
        </VBox>

        <VBox spacing="4">
            <Label text="Xac Nhan Mat Khau" styleClass="field-label"/>
            <PasswordField fx:id="txtConfirm" promptText="nhap lai mat khau"
styleClass="field"/>
        </VBox>

        <VBox spacing="4">
            <Label text="Email (khong bat buoc)" styleClass="field-label"/>
            <TextField fx:id="txtEmail" promptText="email@example.com"
styleClass="field"/>
        </VBox>

        <VBox spacing="4">

```

```

        <Label text="Role" styleClass="field-label"/>
        <ComboBox fx:id="cbRole" maxWidth="Infinity" styleClass="field"/>
    </VBox>

    <Button text="Dang Ky" onAction="#handleRegister"
            styleClass="btn-primary" maxWidth="Infinity"
defaultButton="true"/>

    <Label fx:id="lblMessage" wrapText="true" alignment="CENTER"
maxWidth="Infinity"/>

    <Separator opacity="0.3"/>

    <Button text="Quay Ve Dang Nhap" onAction="#goLogin"
            styleClass="btn-secondary" maxWidth="Infinity"/>
</VBox>
</AnchorPane>

```

dashboard.fxml

```

<?xml version="1.0" encoding="UTF-8"?>
<?import javafx.scene.layout.*?>
<?import javafx.scene.control.*?>
<?import javafx.geometry.*?>

<BorderPane prefHeight="520.0" prefWidth="720.0"
    xmlns:fx="http://javafx.com/fxml/1"
    fx:controller="com.projectmanager.ui.controllers.DashboardController"
    stylesheets="@../styles/main.css">

    <!-- TOP: Header với welcome + logout -->
    <top>
        <HBox alignment="CENTER_LEFT" spacing="16" styleClass="header">
            <Label text="Project Manager" styleClass="title"/>
            <Region HBox.hgrow="ALWAYS"/>
            <VBox alignment="CENTER_RIGHT" spacing="2">
                <Label fx:id="lblWelcome" styleClass="subtitle"/>
                <Label fx:id="lblRole" styleClass="field-label"/>
            </VBox>
            <Button text="Dang Xuat" onAction="#handleLogout" styleClass="btn-
danger"/>
        </HBox>
    </top>

    <!-- LEFT: Sidebar điều hướng -->
    <left>
        <VBox spacing="8" styleClass="sidebar" prefWidth="200">
            <padding><Insets top="20" bottom="20" left="12" right="12"/></padding>
            <Label text="Dieu Huong" styleClass="field-label"/>
            <Separator opacity="0.3"/>

```

```

        <Button text="Them Task"      onAction="#goAddTask"
styleClass="sidebar-btn" maxWidth="Infinity"/>
        <Button text="Danh Sach Task" onAction="#goTaskList"
styleClass="sidebar-btn" maxWidth="Infinity"/>
        <Separator opacity="0.3"/>
        <!-- btnManageUsers: ẩn với user thường (setVisible/setManaged = false
trong Controller) -->
        <Button fx:id="btnManageUsers" text="Quan Ly Users"
            onAction="#goManageUsers" styleClass="sidebar-btn-admin"
maxWidth="Infinity"/>
        </VBox>
    </left>

    <!-- CENTER: Thống kê tổng quan -->
    <center>
        <VBox alignment="CENTER" spacing="24">
            <padding><Insets top="40" left="40" right="40"/></padding>
            <Label text="Tong Quan Du An" styleClass="title"/>
            <Label fx:id="lblStats" styleClass="stats-label" wrapText="true"
                alignment="CENTER" maxWidth="400"/>
            <Label text="Chon menu ben trai de bat dau."
                styleClass="field-label" wrapText="true"/>
        </VBox>
    </center>
</BorderPane>

```

add_task.fxml

```

<?xml version="1.0" encoding="UTF-8"?>
<?import javafx.scene.layout.*?>
<?import javafx.scene.control.*?>
<?import javafx.geometry.*?>

<BorderPane prefHeight="620.0" prefWidth="720.0"
    xmlns:fx="http://javafx.com/fxml/1"
    fx:controller="com.projectmanager.ui.controllers.AddTaskController"
    stylesheets="@../styles/main.css">

    <top>
        <HBox alignment="CENTER_LEFT" spacing="12" styleClass="header">
            <Button text="← Quay Lai" onAction="#goBack" styleClass="btn-
secondary"/>
            <Label text="Them Task Moi" styleClass="title"/>
        </HBox>
    </top>

    <center>
        <ScrollPane fitToWidth="true" styleClass="scroll-pane">
            <VBox spacing="16" styleClass="form-container">
                <padding><Insets top="24" bottom="24" left="40" right="40"/>

```

```

</padding>

    <!-- Chọn loại task -->
    <HBox alignment="CENTER_LEFT" spacing="16">
        <Label text="Loai Task" styleClass="field-label"
prefWidth="160"/>
        <HBox spacing="8">
            <ToggleButton fx:id="btnBug" text="Bug"
styleClass="toggle-btn"/>
            <ToggleButton fx:id="btnFeature" text="Feature"
styleClass="toggle-btn"/>
        </HBox>
    </HBox>

    <!-- Task ID -->
    <HBox alignment="CENTER_LEFT" spacing="16">
        <Label text="Task ID (VD: B001)" styleClass="field-label"
prefWidth="160"/>
        <TextField fx:id="txtId" promptText="B001 hoac F001"
styleClass="field" HBox.hgrow="ALWAYS"/>
    </HBox>

    <!-- Tiêu đề -->
    <HBox alignment="CENTER_LEFT" spacing="16">
        <Label text="Tieu De" styleClass="field-label"
prefWidth="160"/>
        <TextField fx:id="txtTitle" promptText="Mo ta ngan gon"
styleClass="field" HBox.hgrow="ALWAYS"/>
    </HBox>

    <!-- Priority -->
    <HBox alignment="CENTER_LEFT" spacing="16">
        <Label text="Do Uu Tien" styleClass="field-label"
prefWidth="160"/>
        <ComboBox fx:id="cbPriority" maxWidth="Infinity"
styleClass="field" HBox.hgrow="ALWAYS"/>
    </HBox>

    <!-- Status -->
    <HBox alignment="CENTER_LEFT" spacing="16">
        <Label text="Trang Thai" styleClass="field-label"
prefWidth="160"/>
        <ComboBox fx:id="cbStatus" maxWidth="Infinity"
styleClass="field" HBox.hgrow="ALWAYS"/>
    </HBox>

    <!-- Bug: Severity (ẩn khi chọn Feature) -->
    <HBox fx:id="rowSeverity" alignment="CENTER_LEFT" spacing="16">
        <Label text="Severity (Bug)" styleClass="field-label"
prefWidth="160"/>
        <ComboBox fx:id="cbSeverity" maxWidth="Infinity"
styleClass="field" HBox.hgrow="ALWAYS"/>
    </HBox>

```

```

        <!-- Feature: Estimated Hours (ẩn khi chọn Bug) -->
        <HBox fx:id="rowHours" alignment="CENTER_LEFT" spacing="16">
            <Label text="Số Giờ Ước Tính" styleClass="field-label"
prefWidth="160"/>
            <TextField fx:id="txtHours" promptText="VD: 8"
styleClass="field" HBox.hgrow="ALWAYS"/>
        </HBox>

        <!-- Feature: Developer (ẩn khi chọn Bug) -->
        <HBox fx:id="rowDeveloper" alignment="CENTER_LEFT" spacing="16">
            <Label text="Giao Cho Developer" styleClass="field-label"
prefWidth="160"/>
            <TextField fx:id="txtDeveloper" promptText="Tên developer (co
the bo trong)"
styleClass="field" HBox.hgrow="ALWAYS"/>
        </HBox>

        <Separator opacity="0.3"/>

        <Button text="Them Task" onAction="#handleAdd"
styleClass="btn-primary" maxWidth="Infinity"
defaultButton="true"/>

        <Label fx:id="lblMessage" wrapText="true" alignment="CENTER"
maxWidth="Infinity"/>
    </VBox>
</ScrollPane>
</center>
</BorderPane>

```

task_list.fxml

```

<?xml version="1.0" encoding="UTF-8"?>
<?import javafx.scene.layout.*?>
<?import javafx.scene.control.*?>
<?import javafx.geometry.*?>

<BorderPane prefHeight="700.0" prefWidth="1050.0"
xmlns:fx="http://javafx.com/fxml/1"
fx:controller="com.projectmanager.ui.controllers.TaskListController"
stylesheets="@../styles/main.css">

    <top>
        <HBox alignment="CENTER_LEFT" spacing="12" styleClass="header">
            <Button text="← Quay Lại" onAction="#goBack" styleClass="btn-
secondary"/>
            <Label text="Danh Sach Task" styleClass="title"/>
        </HBox>
    </top>

```

```

<center>
    <VBox spacing="12">
        <padding><Insets top="16" bottom="16" left="20" right="20"/></padding>

        <!-- Toolbar: Filter + Update status + Delete -->
        <HBox spacing="8" alignment="CENTER_LEFT">
            <Label text="Status:" styleClass="field-label"/>
            <ComboBox fx:id="cbFilterStatus" styleClass="field"
prefWidth="130"/>
            <Label text="Loai:" styleClass="field-label"/>
            <ComboBox fx:id="cbFilterType" styleClass="field"
prefWidth="100"/>
            <Button text="Loc" onAction="#handleFilter"
styleClass="btn-primary"/>
            <Button text="Xoa Loc" onAction="#handleClearFilter"
styleClass="btn-secondary"/>
            <Region HBox.hgrow="ALWAYS"/>
            <Label text="Doi Status:" styleClass="field-label"/>
            <ComboBox fx:id="cbNewStatus" styleClass="field" prefWidth="130"/>
            <Button text="Cap Nhat" onAction="#handleUpdateStatus"
styleClass="btn-secondary"/>
            <!-- btnDelete: ấn với user thường, hiện với admin (set trong
Controller) -->
            <Button fx:id="btnDelete" text="Xoa Task" onAction="#handleDelete"
styleClass="btn-danger"/>
            <Button text="Tai Lai DB" onAction="#handleReload"
styleClass="btn-secondary"/>
        </HBox>

        <!-- TableView -->
        <TableView fx:id="tableView" VBox.vgrow="ALWAYS" styleClass="table-
view">
            <columns>
                <TableColumn fx:id="colType" text="Loai"
prefWidth="60"/>
                <TableColumn fx:id="colId" text="ID"
prefWidth="80"/>
                <TableColumn fx:id="colTitle" text="Tieu De"
prefWidth="260"/>
                <TableColumn fx:id="colPriority" text="Priority"
prefWidth="90"/>
                <TableColumn fx:id="colStatus" text="Status"
prefWidth="120"/>
                <TableColumn fx:id="colEffort" text="Effort"
prefWidth="75"/>
                <TableColumn fx:id="colExtra" text="Severity / Developer"
prefWidth="200"/>
            </columns>
            <placeholder>
                <Label text="Khong co task nao." styleClass="field-label"/>
            </placeholder>
        </TableView>

        <!-- Stats bar -->

```

```

        <HBox spacing="40" alignment="CENTER_LEFT">
            <Label fx:id="lblEffort" styleClass="subtitle"/>
            <Label fx:id="lblStatus" styleClass="field-label"/>
        </HBox>
    </VBox>
</center>
</BorderPane>

```

user_list.fxml

```

<?xml version="1.0" encoding="UTF-8"?>
<?import javafx.scene.layout.*?>
<?import javafx.scene.control.*?>
<?import javafx.geometry.*?>

<BorderPane prefHeight="580.0" prefWidth="750.0"
    xmlns:fx="http://javafx.com/fxml/1"
    fx:controller="com.projectmanager.ui.controllers.UserListController"
    stylesheets="@../styles/main.css">

    <top>
        <HBox alignment="CENTER_LEFT" spacing="12" styleClass="header">
            <Button text="← Quay Lai" onAction="#goBack" styleClass="btn-
secondary"/>
            <Label text="Quan Ly Users (Admin)" styleClass="title"/>
            <Region HBox.hgrow="ALWAYS"/>
            <Label text="[ADMIN ONLY]" styleClass="tag-admin"/>
        </HBox>
    </top>

    <center>
        <VBox spacing="12">
            <padding><Insets top="16" bottom="16" left="20" right="20"/></padding>

            <!-- Toolbar -->
            <HBox spacing="8" alignment="CENTER_LEFT">
                <Button text="Khoa / Mo Khoa" onAction="#handleToggleStatus"
styleClass="btn-warning"/>
                <Button text="Lam Moi" onAction="#handleRefresh"
styleClass="btn-secondary"/>
                <Region HBox.hgrow="ALWAYS"/>
                <Label fx:id="lblMessage" styleClass="field-label"/>
            </HBox>

            <!-- TableView -->
            <TableView fx:id="tableView" VBox.vgrow="ALWAYS" styleClass="table-
view">
                <columns>
                    <TableColumn fx:id="colId" text="ID"
prefWidth="60"/>

```



```

        <TableColumn fx:id="colUsername" text="Username"
prefWidth="150"/>
        <TableColumn fx:id="colEmail" text="Email"
prefWidth="220"/>
        <TableColumn fx:id="colRole" text="Role"
prefWidth="90"/>
        <TableColumn fx:id="colStatus" text="Trang Thai"
prefWidth="100"/>
    </columns>
    <placeholder>
        <Label text="Khong co user nao." styleClass="field-label"/>
    </placeholder>
</TableView>
</VBox>
</center>
</BorderPane>

```

main.css

```

/* — Root ————— */

.root {
    -fx-background-color: #1e1e2e;
    -fx-font-family: Georgia, serif;
    -fx-font-size: 13px;
}

/* — Typography ————— */

.title {
    -fx-text-fill: #cdd6f4;
    -fx-font-size: 22px;
    -fx-font-weight: bold;
}

.subtitle {
    -fx-text-fill: #a6adc8;
    -fx-font-size: 13px;
}

.field-label {
    -fx-text-fill: #6c7086;
    -fx-font-size: 11px;
}

.stats-label {
    -fx-text-fill: #cdd6f4;
    -fx-font-size: 14px;
    -fx-line-spacing: 4;
}

```

```
.label {
    -fx-text-fill: #cdd6f4;
}

/* — Form Fields — */

.field {
    -fx-background-color: #313244;
    -fx-text-fill: #cdd6f4;
    -fx-prompt-text-fill: #585b70;
    -fx-border-color: #45475a;
    -fx-border-radius: 5;
    -fx-background-radius: 5;
    -fx-padding: 8 12;
}

.field:focused {
    -fx-border-color: #89b4fa;
    -fx-background-color: #363649;
}

/* — Buttons — */

.btn-primary {
    -fx-background-color: #89b4fa;
    -fx-text-fill: #1e1e2e;
    -fx-font-weight: bold;
    -fx-border-radius: 5;
    -fx-background-radius: 5;
    -fx-padding: 9 18;
    -fx-cursor: hand;
}

.btn-primary:hover { -fx-background-color: #b4befe; }
.btn-primary:pressed { -fx-background-color: #7287fd; }

.btn-secondary {
    -fx-background-color: transparent;
    -fx-text-fill: #89b4fa;
    -fx-border-color: #45475a;
    -fx-border-radius: 5;
    -fx-background-radius: 5;
    -fx-padding: 9 18;
    -fx-cursor: hand;
}

.btn-secondary:hover { -fx-background-color: #313244; -fx-border-color: #89b4fa; }

.btn-danger {
    -fx-background-color: #f38ba8;
    -fx-text-fill: #1e1e2e;
    -fx-font-weight: bold;
    -fx-border-radius: 5;
    -fx-background-radius: 5;
    -fx-padding: 9 18;
}
```

```
-fx-cursor: hand;
}
.btn-danger:hover { -fx-background-color: #eba0ac; }

/* Nút vàng – dùng cho block/unblock user (nguy hiểm nhưng không destructive như
xóa) */
.btn-warning {
    -fx-background-color: #fab387;
    -fx-text-fill: #1e1e2e;
    -fx-font-weight: bold;
    -fx-border-radius: 5;
    -fx-background-radius: 5;
    -fx-padding: 9 18;
    -fx-cursor: hand;
}
.btn-warning:hover { -fx-background-color: #f9e2af; }

/* — Toggle Buttons ————— */

.toggle-btn {
    -fx-background-color: #313244;
    -fx-text-fill: #a6adc8;
    -fx-border-color: #45475a;
    -fx-border-radius: 5;
    -fx-background-radius: 5;
    -fx-padding: 8 24;
    -fx-cursor: hand;
}
.toggle-btn:selected {
    -fx-background-color: #89b4fa;
    -fx-text-fill: #1e1e2e;
    -fx-font-weight: bold;
    -fx-border-color: #89b4fa;
}

/* — Layout ————— */

.header {
    -fx-background-color: #181825;
    -fx-padding: 12 20;
    -fx-border-color: #45475a;
    -fx-border-width: 0 0 1 0;
}

.sidebar {
    -fx-background-color: #181825;
    -fx-padding: 20 12;
    -fx-border-color: #45475a;
    -fx-border-width: 0 1 0 0;
}

.sidebar-btn {
    -fx-background-color: transparent;
    -fx-text-fill: #a6adc8;
```

```

    -fx-alignment: CENTER_LEFT;
    -fx-padding: 10 16;
    -fx-cursor: hand;
    -fx-border-radius: 5;
    -fx-background-radius: 5;
}
.sidebar-btn:hover { -fx-background-color: #313244; -fx-text-fill: #cdd6f4; }

/* Sidebar button admin – màu vàng cam để phân biệt */
.sidebar-btn-admin {
    -fx-background-color: transparent;
    -fx-text-fill: #fab387;
    -fx-alignment: CENTER_LEFT;
    -fx-padding: 10 16;
    -fx-cursor: hand;
    -fx-border-radius: 5;
    -fx-background-radius: 5;
}
.sidebar-btn-admin:hover { -fx-background-color: #313244; -fx-text-fill: #f9e2af;
}

.form-container { -fx-background-color: #1e1e2e; }
.scroll-pane { -fx-background-color: #1e1e2e; -fx-border-color: transparent; }
.scroll-pane > .viewport { -fx-background-color: #1e1e2e; }

/* Tag "ADMIN ONLY" trong header user_list */
.tag-admin {
    -fx-text-fill: #f38ba8;
    -fx-font-size: 11px;
    -fx-font-weight: bold;
    -fx-background-color: #3b1f2b;
    -fx-padding: 3 8;
    -fx-background-radius: 4;
}

/* — TableView ————— */

.table-view {
    -fx-background-color: #313244;
    -fx-border-color: #45475a;
    -fx-border-radius: 5;
    -fx-table-cell-border-color: #45475a;
}
.table-view .column-header-background { -fx-background-color: #45475a; }
.table-view .column-header .label {
    -fx-text-fill: #cdd6f4;
    -fx-font-weight: bold;
    -fx-font-size: 12px;
}
.table-view .table-row-cell {
    -fx-background-color: #313244;
    -fx-border-color: #45475a;
    -fx-border-width: 0 0 1 0;
}

```

```

.table-view .table-row-cell:selected { -fx-background-color: #45475a; }
.table-view .table-row-cell:hover    { -fx-background-color: #363649; }
.table-view .table-cell {
    -fx-text-fill: #cdd6f4;
    -fx-font-size: 12px;
    -fx-padding: 6 8;
}

/* — Separator & ComboBox ————— */

.separator .line { -fx-border-color: #45475a; -fx-border-width: 1 0 0 0; }

.combo-box {
    -fx-background-color: #313244;
    -fx-text-fill: #cdd6f4;
    -fx-border-color: #45475a;
    -fx-border-radius: 5;
    -fx-background-radius: 5;
}
.combo-box .list-cell          { -fx-text-fill: #cdd6f4; -fx-background-color:
#313244; }
.combo-box-popup .list-view { -fx-background-color: #313244; -fx-border-color:
#45475a; }
.combo-box-popup .list-view .list-cell:hover { -fx-background-color: #45475a; }

```

Ghi Chú FXML & CSS

Điểm	Giải thích
stylesheets="@../styles/main.css"	@ prefix = đường dẫn tương đối — từ <code>views/</code> lên <code>ui/</code> rồi vào <code>styles/</code>
fx:controller="com.projectmanager..."	JavaFX tự inject @FXML fields từ controller class
defaultButton="true"	Enter phím phản hồi nút Login/Đăng ký
HBox.hgrow="ALWAYS"	TextField co giãn chiếm hết không gian ngang còn lại trong HBox
VBox.vgrow="ALWAYS"	TableView co giãn chiếm hết chiều cao trong VBox
prefWidth="160" trên Label form	Căn chỉnh cột label — mọi field cùng vị trí ngang
fx:id="btnManageUsers" với setManaged	Controller gọi <code>setManaged(false)</code> khi ẩn — nút không chiếm chỗ trong VBox layout
tag-admin CSS class	Label "[ADMIN ONLY]" trong user_list — dùng <code>background-color</code> + <code>border-radius</code> để bo
btn-warning màu cam	Phân biệt với <code>btn-danger</code> đỏ — block user nguy hiểm hơn thao tác thường nhưng ít hơn xóa
Dark theme #1E1E2E	Background chính — Catppuccin Mocha palette

Điểm	Giải thích
#F38BA8 Bug / #89DCEB Feature	Màu đỏ hồng (Bug) và teal (Feature) trong TableView cột Type
#F38BA8/#FAB387/#A6E3A1 Priority	Đỏ (HIGH), Cam (MEDIUM), Xanh lá (LOW)
sidebar-btn-admin cam nhạt	Phân biệt nút admin với nút thường trong sidebar — dễ nhận ra quyền hạn